

Orthopolity: a general law of resource allocation

Effective resources, constrained distributions, and inverse inference

Research manuscript, revised 6 October 2026. The analyses of Sections 3 and 4 are exploratory or governed by a locally recorded plan. Section 5 distinguishes forecasts frozen before outcome decoding from a plant transfer study frozen after raw outcome exposure. None is an externally preregistered study. Authorship and submission declarations require agreement before submission. Historical drafts remain in Git history. Computational details accompany the manuscript in `evidence.md` and `source-audit.md`.

Abstract

We propose orthopolity as a new general natural law: Nature tends to distribute resources equally among concentrations of those resources. Its expression depends on the physical comparison measure, interactions, and system and medium constraints. The relevant resource may be an effective combination of several resources, including nonlinear combinations; equality in an individual measured resource is therefore only one possible manifestation. We formulate the law's neutral allocation relation and its constrained extensions, and develop both a forward problem, predicting abundance from resources, and an inverse problem, inferring effective resource structure from observed distributions. Under logarithmic size allocation, a count density proportional to $k^{-(1+\alpha)}$ identifies effective resource divided by the constraint factor as proportional to k^α . Multiple environments can distinguish resource compositions that a single distribution cannot. A class-exchange mechanism proves equalization, an unequal equilibrium with opposing active contributions, and recovery after constraint removal. Electromagnetic transport and thermal radiation supply established physical realizations with explicit comparison measures and quantitative departures. A retrospective FIRAS calculation connects mode allocation to a measured spectrum while preserving the product's reconstruction and uncertainty assumptions. Retained empirical studies provide positive conditional budget-closure forecasts, failed redistribution forecasts, and unresolved natural size-class neutrality. City-size scaling illustrates an additional inverse application. The contribution is the general-law proposal and its mathematical, physical, and inferential framework; the component derivations and observations have their stated established precedents. The results support particular realizations and predictions; extension of the law's scope and identification of effective resources remain empirical tasks.

1. Introduction

We propose **orthopolity as a general natural law** governing the allocation of resources among their concentrations. Its central statement is:

Nature tends to distribute resources equally among concentrations of those resources.

The law concerns a physical tendency whose observed expression depends on the system. Equal resource allocation can require unequal numbers of objects: concentrations that require more effective resource can be less abundant. This article develops the proposal's basis, mathematical consequences, mechanisms, physical realizations, empirical evidence, and inverse use for identifying

the resources underlying observed systems. The general statement is a proposed law; each worked result specifies the conditions under which it has been derived or observed.

The term follows Fabbri's 2024 essay and an earlier manuscript by Fabbri and Oliveira Jr. (2017). The earlier account assumes uniform resource allocation across component-wealth values; the later essay proposes a broader natural and cosmological principle. Here we develop a common formulation while keeping each physical comparison explicit.

Real concentrations depend on several resources together. Algae develop within available space, light, water and multiple nutrients over time; cities depend on land, people, buildings, transport, energy, money, food and water. These quantities interact and may substitute for, complement, or limit one another. The operative resource may therefore be a nonlinear effective combination. Deriving it from individual inputs can be difficult. Orthopolity also motivates the reverse direction: use an observed allocation to infer properties of the effective resource, then test the resulting resource model through independent measurements or predictions in other conditions. Forward prediction and inverse inference are complementary uses of the law, developed in Sections 2.2–2.3.

The proposed tendency must be distinguished from the resulting distribution. A standing tree experiences gravity while its structure and anchorage support it. An apple thrown upward initially rises while gravity acts; catching it or lodging it against a ceiling introduces an interaction that can prevent its return to the ground. These outcomes reflect gravity together with other interactions and initial conditions. The same logical distinction guides orthopolity: its visible signature can be altered or obscured by the construction of the system and its medium. Human age and height distributions motivate analyses of such constraints. Their lack of an apparent neutral signature would not, by itself, establish a general absence of evidence for orthopolity in Nature. Section 2.7 states how this distinction enters the research programme.

The ecological precedent is substantial. Sheldon, Prakash and Sutcliffe (1972) studied ocean particle size distributions; Gaedke (1993) explicitly related equal biomass in logarithmic classes to normalized biomass spectra; Hatton et al. (2021) reconstructed the ocean spectrum from bacteria to whales. Arranz et al. (2022) studied variation among hundreds of lake fish communities and its ecological correlates. Variation among individual systems is therefore already a research topic. Energetic equivalence (Damuth, 1981) is related, but its species-population sampling unit differs from community abundance per size interval. Species richness across sizes prevents automatic interchange of those statements.

Power laws arise under different generative mechanisms, so an observed exponent supplies an inverse constraint rather than a unique causal explanation (Newman, 2005; Clauset, Shalizi and Newman, 2009). Our formulation specifies what a distribution can reveal about effective resources and what additional information resolves competing interpretations. The class-exchange model demonstrates an equalizing contribution that persists under a constraint. Electromagnetic transport and thermal radiation connect the proposed law to known physical equations, with different physically defined comparison measures. The empirical cases establish the successes and limits of specific resource and response models. The accounting identity, reversible transport, classical equipartition and quantum radiation law are established results; their organization into a general allocation proposal and its forward and inverse programme is the contribution developed here.

2. Mathematical formulation

2.1 Objects, resources, and a reference measure

Let k be a positive size coordinate on a finite domain $D = [a, b]$, with $0 < a < b < \infty$. Choose a reference measure μ , with $0 < \mu(D) < \infty$, for the classes being compared. Let $n_\mu(k) = dN/d\mu$ denote object abundance and $\bar{q}(k) = E[q \mid k] > 0$ the conditional arithmetic mean of an additive resource per object. Then

$$\mathcal{O}_\mu(k) = \frac{dR}{d\mu} = \bar{q}(k)n_\mu(k). \quad (1)$$

For an observed class, the corresponding identity is simply $\sum_i q_i = N_{\text{class}}\bar{q}_{\text{class}}$. Resources may be a standing stock, such as biomass, or an accumulated event quantity over a declared observation period. The latter is not a stock resident in a size class, and neither is a flux through size classes. Additivity is needed for (1); a dynamical conservation law is not.

Define the **neutral allocation form of orthopolity relative to μ** , denoted $\mathcal{O}_\mu$, by

$$\mathcal{O}_\mu(k) = C > 0 \quad \text{for } \mu\text{-almost every } k \in D. \quad (2)$$

This is an assumption about the neutral allocation, rather than a claim that every constrained system has this profile. It gives $n_\mu = C/\bar{q}$. Equivalently, if an object is sampled uniformly from the population, its size density with respect to μ is

$$p_N(k) = \frac{1}{Z\bar{q}(k)}, \quad Z = \int_D \frac{d\mu(k)}{\bar{q}(k)} < \infty. \quad (3)$$

Sampling instead in proportion to the resource held gives $p_R(k) = \bar{q}(k)p_N(k)/E_N[q] = 1/\mu(D)$. Thus uniform resource-weighted size and inverse-cost object abundance are equivalent statements, under the same reference measure. This is an accounting equivalence, not a derivation of the allocation assumption. The count, resource total, and constants also satisfy $N = CZ$ and $R = C\mu(D)$. For a finite observed population, the identity is exact for class totals; a smooth density is a population model or approximation.

2.2 Effective resources in systems with coupled requirements

An effective resource can combine several constituent requirements. For an object i , let $q_{i\ell}$ be its requirement or assigned amount of constituent ℓ , and let $s_\ell > 0$ be a fixed reference quantity in the same units. A general representation is

$$Q_i = Q_0 F\left(\frac{q_{i1}}{s_1}, \dots, \frac{q_{im}}{s_m}; z_i, \theta\right) > 0, \quad \bar{q}_{\text{eff}}(k) = E[Q_i \mid k].$$

Here Q_0 supplies effective-resource units, z_i records relevant environmental or historical conditions, and θ specifies the combination rule. The reference scales are declared independently of the observed abundance. Nonlinearity can represent complementarity, substitution, saturation, or a bottleneck. The arithmetic class mean is taken after combining the constituent resources. In general, $E[F(X, Y) \mid k] \neq F(E[X \mid k], E[Y \mid k])$; for $F(X, Y) = XY$, the difference is the conditional covariance. Marginal resource means alone may therefore miss a dependence that changes the effective resource.

The quantity $R_{\text{eff}} = \sum_i Q_i$ is additive over the declared object partition, so equation (1) still holds with $\bar{q} = \bar{q}_{\text{eff}}$. Nonlinearity within an object does not prevent this additivity across objects. It does, however, make the physical partition important: $\sum_i F(\mathbf{q}_i)$ need not equal $F(\sum_i \mathbf{q}_i)$.

Splitting or merging objects can change the effective total. Nor is R_{eff} automatically conserved merely because some constituent resources are conserved. The proposed combination must be interpreted as a physical stock, an accumulated requirement, an opportunity cost, or another explicitly defined quantity, with its appropriate balance law. Collective interaction terms must be included or assigned to objects through a physical accounting convention without double counting.

For algae, nutrient stocks per cell and available nutrient budgets differ; light can be an absorbed photon flux or an exposure integrated over a specified interval. Space and water can be capacities or environmental conditions. Time can enter through residence time, growth history, or an additive occupancy such as cell-hours; elapsed time is not automatically a divisible stock. For cities, land, buildings and people are stocks, while food, water, energy and money can involve stocks, flows and exchanges over the chosen observation horizon. The combination rule must respect those distinctions and avoid counting a resource twice along a supply chain.

Multiple physical budgets remain explicit. For fixed class requirements, feasibility can require $\sum_j N_j q_{\ell j} \leq B_\ell$ for each constituent ℓ . A scalar Q does not generally replace all these inequalities. A specified optimization model can produce a local effective cost $Q_i = \sum_\ell \lambda_\ell q_{i\ell}$, with dimensionally compatible scarcity prices λ_ℓ , but those prices and their range of validity belong to that model (Boyd and Vandenberghe, 2004). A bottleneck gives another illustration: for a separately supported population of identical objects with fixed requirements, $N_{\text{max}} = \min_\ell B_\ell / q_\ell$ depends nonlinearly on the budgets. It does not prescribe the allocation among competing types. These are ways to derive candidate effective quantities; neither scalarization nor feasibility by itself proves neutrality.

A simple nonlinear candidate is a product of positive normalized inputs, $Q = Q_0 \prod_\ell (q_\ell / s_\ell)^{\theta_\ell}$. If constituent requirements are deterministic functions $q_\ell(k) \propto k^{d_\ell}$, then $Q(k) \propto k^{d_{\text{eff}}}$ with $d_{\text{eff}} = \sum_\ell \theta_\ell d_\ell$. Under logarithmic neutrality the abundance density exponent is $1 + d_{\text{eff}}$. For differentiable F , the more general local elasticity is

$$\frac{d \ln Q}{d \ln k} = \sum_\ell \frac{\partial \ln F}{\partial \ln(q_\ell / s_\ell)} \frac{d \ln q_\ell}{d \ln k},$$

when the environment and parameters are fixed and F has no further explicit size dependence. Varying constituent elasticities or interactions can generate curvature. This chain rule for specified functions is distinct from a claim that independent random resource distributions have additive tail exponents. With heterogeneous objects, the class mean $E[Q | k]$ remains the relevant quantity. The effective-resource derivation develops the finite-bin and heterogeneity implications.

2.3 Inverse inference and the city-size example

The law can be used in two directions. Given a resource model and a specified constraint, it predicts abundance. Given an abundance profile, it supplies information about the effective resource required by that allocation. Write a positive constrained profile as

$$\bar{q}_{\text{eff}}(k) n_\mu(k) = C a(k), \quad a(k) > 0,$$

where neutrality has $a = 1$. The class-exchange example below derives $a = e^{-V}$ from prescribed kinetics. In other systems a needs its own mechanism or conditional assumption; an arbitrary fitted factor is not an explanation. On a domain with positive population density, inversion gives

$$\frac{\bar{q}_{\text{eff}}(k)}{a(k)} = \frac{C}{n_\mu(k)}, \quad \frac{\bar{q}_{\text{eff}}(k)/a(k)}{\bar{q}_{\text{eff}}(k_0)/a(k_0)} = \frac{n_\mu(k_0)}{n_\mu(k)}.$$

Thus a distribution identifies the shape of effective resource divided by its constraint factor, conditional on the comparison measure. With the constraint profile and its normalization specified, an independently measured effective-resource total fixes $C = R_{\text{eff}} / \int_D a d\mu$. Without further information, the constituents and their combination are not unique; multiplying both $\bar{q}_{\text{eff}}$ and a by the same positive size-dependent function leaves the prediction unchanged. This is a precise statement of the inverse information available, not a reason to discard inverse inference. Measured constituents, restricted combination families, and responses in different environments can resolve parts of that ambiguity. Finite observed counts require a sampling model and bin integration; a zero-count bin does not establish infinite physical cost.

A constructive inverse example. Prespecify $Q = Q_0 X^{\theta_1} Y^{\theta_2}$ for positive normalized resources. In environment s , suppose independently measured class requirements scale as $X_s \propto k^{d_{1s}}$ and $Y_s \propto k^{d_{2s}}$, with no within-class variation, and a known constraint has $V_s = v_s \ln(k/k_0)$. A Pareto count density $dN/dk \propto k^{-(1+\alpha_s)}$ then implies

$$\alpha_s - v_s = \theta_1 d_{1s} + \theta_2 d_{2s}.$$

Two environments with scaling vectors $(1, 1)$ and $(1, 2)$ and corrected exponents $\alpha_s - v_s$ equal to 1.5 and 2 identify $\theta_1 = 1$ and $\theta_2 = 1/2$. The inferred nonlinear resource is $Q = Q_0 X \sqrt{Y}$. A third environment with scaling vector $(2, 1)$ predicts $\alpha_3 = 2.5 + v_3$ without refitting. The two independent scaling directions identify the composition within this family; the third supplies a test. The remaining scale Q_0 requires an independent anchor. These are algebraic illustrative values, not measurements. They show how allocation distributions can reveal a composite resource and generate further predictions.

City sizes. Upper city-size tails often admit Pareto descriptions, with Zipf scaling as a prominent special case. The result depends on how cities and the fitted size range are defined. Eeckhout (2004) found a lognormal description across the full US Census Places distribution; Ioannides and Skouras (2013) found a lognormal body and Pareto tail, with an estimated tail exponent of 1.25 for their preferred Census Places model. These findings concern different parts and definitions of an urban system, not a requirement that every city-size observation follow exact Zipf scaling.

Let S denote population per city and $n(S)$ the positive number density of cities per unit population interval. For a Pareto survival exponent α , $N(\geq S) \propto S^{-\alpha}$ and $n(S) \propto S^{-(1+\alpha)}$; ranked size scales as $S(r) \propto r^{-1/\alpha}$. Finite upper cutoffs modify the survival and rank relations near the largest city. Under logarithmic allocation,

$$\frac{dN}{d \ln S} = S n(S), \quad \frac{\bar{q}_{\text{eff}}(S)}{a(S)} \propto S^\alpha.$$

Exact Zipf scaling has $\alpha = 1$. If $a = 1$ and population is the resource, $q(S) = S$, the relation expresses equal population in equal logarithmic city-size intervals. If the operative resource is a nonlinear combination of urban requirements, the same observation instead constrains that combination to have approximately linear net scaling, after accounting for a . This is the proposed orthopolity interpretation and an inverse target for measurements of land, infrastructure, energy, water and other inputs. The exponent alone does not identify the combination or assign a probability to its causal explanation. Existing proportional-growth mechanisms, including Gabaix's (1999) model with lower-size regularization, provide comparisons that the resource account should explain, connect to, or distinguish. No new city dataset is analyzed here.

2.4 The measure changes the exponent

Write $u = \ln(k/k_0)$ for a fixed reference size k_0 . For $\bar{q}(k) = q_0(k/k_0)^d$, with $d > 0$, two different hypotheses give:

Equal resource with respect to	Resource spectrum	Count density with respect to dk
Linear size, $d\mu = dk$	$\bar{q}(k)dN/dk = C$	$dN/dk \propto k^{-d}$
Logarithmic size, $d\mu = du = dk/k$	$\bar{q}(k)dN/du = C$	$dN/dk \propto k^{-(d+1)}$

The second row follows because $dN/du = k dN/dk$. For biomass with $q = k = m$, the predictions are consequently density exponents 1 and 2, respectively. The 2017 manuscript's explicit uniform density over a linear wealth interval is not interchangeable with the logarithmic convention used in the empirical analyses below. Adopting logarithmic classes is a substantive specification, motivated by size-spectrum practice.

Logarithmic allocation is unchanged by changing size units or logarithm base, apart from a constant density multiplier. It also survives $z = (k/k_0)^c$ for constant $c > 0$, since $d \ln z = c d \ln k$. It is not invariant under arbitrary nonlinear coordinates. In general, $dR/dv = (dR/du)|du/dv|$. Choosing a transformation after observing R can therefore manufacture flatness; the coordinate and measure must be fixed independently.

Under $O_{\log}$ and the power-cost assumption, the slope of abundance per log interval is $-d$, the ordinary density exponent is $d + 1$, and the survival function on $[a, b]$ is

$$P(K \geq k) = \frac{k^{-d} - b^{-d}}{a^{-d} - b^{-d}}. \quad (4)$$

The familiar survival exponent d and rank-size exponent $1/d$ are approximations away from the upper boundary, not exact finite-domain identities. For an unbounded domain with $d > 0$, count can be finite while logarithmically uniform resource has infinite total. Finite boundaries are essential to a finite-budget interpretation.

2.5 Allocation postulates and limits of the accounting

Resource symmetry as a candidate law. The proposed rationale is that, absent restrictions favouring particular resource costs, allocation should not distinguish those costs. A natural law can adopt this as a physical postulate without a deeper mechanistic derivation. One precise sufficient formulation uses K declared classes with fixed positive per-object costs q_i , allocations $R_i = q_i N_i$, and fixed total $\sum_i R_i = R > 0$. If the probability law of $(R_1, \dots, R_K)$ is invariant under exchanging allocations across the fixed classes, then symmetry and the sum give

$$E[R_i] = R/K, \quad E[N_i] = R/(Kq_i).$$

This postulate selects resource allocations as the symmetric quantities; symmetry of counts would give a different prediction. It is stronger than conservation and is not implied by an unspecified absence of constraints. Indivisible objects may require an approximation where exact exchangeability is infeasible. The class measure also matters: with cost itself as coordinate, equal resource per dq predicts $dE[N]/dq \propto q^{-1}$, whereas equal resource per $d \ln q$ predicts $dE[N]/dq \propto q^{-2}$. Transformation-based treatments of indifference likewise require a specified symmetry (Jaynes, 1968); an inferential rule does not itself establish a physical law.

Equal expectation does not establish equal snapshots or relaxation toward equality. In a divisible-resource idealization, assigning all resource to one uniformly chosen class has the required symmetry and unequal allocation in every realization. Time-average equality needs an additional condition connecting time averages to expectations or its own empirical postulate. A test must therefore specify what "tends to allocate" means and identify eligible systems through conditions

independent of their observed allocation. A deeper mechanism is optional for an empirical law; the scope and predictive evidence are not.

Conservation does not imply equal allocation. On any bounded logarithmic domain, $\mathcal{O}(u) = A \exp(\beta u)$ can be normalized to the same resource total for every β . Conservation fixes the integral and leaves its shape undetermined. Likewise, maximum entropy over finitely many equally weighted object classes, constrained by a mean cost, yields $p_j \propto \exp(-\lambda q_j)$. Different reference measures or constraints give different distributions (Jaynes, 1957; Visser, 2013).

Equipartition does not imply a power law without a cost law. If $\bar{q}(k) = q_0 \exp(k/k_0)$, then $O_{\log}$ gives $dN/du \propto \exp(-k/k_0)$ and $dN/dk \propto \exp(-k/k_0)/k$. Neither is a power law. Geometric dimension supplies d only in a specified geometric model such as similar objects with volume proportional to length cubed. It does not establish the number of independent resource inputs in an arbitrary system.

Independent inputs do not generally add tail exponents. For example, two independent unit-scale Pareto variables with densities hx^{-h-1} , $x \geq 1$, $h > 0$, have product density $h^2 z^{-h-1} \ln z$, $z \geq 1$. Integration over the possible factorizations introduces a logarithmic factor. This limits the independent-input argument in the 2017 essay.

Forward tests and inverse inference have different roles. For any positive abundance, defining $\bar{q} = C/n_\mu$ makes (2) hold identically. In Section 2.3 this is the starting inverse relation: it identifies a conditional effective-resource profile and motivates resource hypotheses. Agreement on the same fitted profile is not independent evidence for the law or for the inferred physical composition. A forward test fixes the resource model through definitions, independent measurements, or a separate calibration, then checks a prediction on additional observations. A constraint inferred from the original profile likewise needs additional information or a transferred prediction to establish its explanatory value.

2.6 What an ensemble statement would mean

Suppose a system has a power-shaped resource spectrum $\mathcal{O}_i(u) = A_i \exp(\beta_i u)$. A proposed ensemble condition $E[\beta_i] = 0$ concerns mean slopes. It does not imply $E[\mathcal{O}_i(u)]$ is constant. With equal $A_i = A$ and $\beta_i \sim N(0, \tau^2)$,

$$E[\mathcal{O}_i(u)] = A \exp(\tau^2 u^2 / 2), \quad (5)$$

which is curved whenever $\tau > 0$. Equal-total normalization changes this formula but does not make mean-zero slopes sufficient: averaging normalized spectra proportional to $\exp(cu)$ and $\exp(-cu)$ on a symmetric domain produces a nonconstant hyperbolic cosine. A median slope is a different estimand again. A claim about average resource occupancy requires spectra, a common domain, and an explicit weighting of systems; slopes alone do not identify it.

More generally, let $H(u) = \sum_{i=1}^n w_i A_i e^{\beta_i u}$ on a common log-domain, with fixed positive weights and amplitudes. Then

$$H''(u) = \sum_i w_i A_i \beta_i^2 e^{\beta_i u} \geq 0.$$

If any slope is nonzero, the derivative is strictly positive, so the arithmetic mixture cannot be constant on an open interval. Exact flatness requires every contributing slope to be zero. With $p_i(u) = w_i A_i e^{\beta_i u} / H(u)$, its log-curvature is $d^2 \ln H / du^2 = \text{Var}_{p(u)}(\beta)$. Per-system total normalization preserves these conclusions. They are standard convexity consequences for exact power spectra; arbitrary curved profiles can cancel to a flat mean. They therefore neither exclude a general resource-symmetry law nor convert noisy fitted slopes into a profile test. The scientific assessment gives a quantitative bound for approximate flatness and its weighting restrictions.

Mean and variance also do not require a Gaussian distribution. Gaussianity is an additional maximum-entropy modelling choice, and selecting it by AIC is not a goodness-of-fit test. If $\beta_i \sim N(\mu_\beta, \tau^2)$ and reported slope error is independent $N(0, \sigma_i^2)$, the predicted probability that the *reported slope* lies in $[-c_i, c_i]$ is

$$\Phi_N\left(\frac{c_i - \mu_\beta}{\sqrt{\tau^2 + \sigma_i^2}}\right) - \Phi_N\left(\frac{-c_i - \mu_\beta}{\sqrt{\tau^2 + \sigma_i^2}}\right). \quad (6)$$

The expression $2\Phi_N(c_i/\tau) - 1$ applies to zero-centred *latent* slopes. Comparing it to observed estimates without their errors is a different calculation. Fitting τ on the same slopes makes either comparison an internal diagnostic, not independent validation.

There is a valid simultaneous-resource restriction for an individual system: $\mathcal{O}_1/\mathcal{O}_2 = \bar{q}_1/\bar{q}_2$. Both spectra can be exactly flat only if their mean resource ratio is size-independent. Across systems, however, $\beta_{1i} - \beta_{2i} = d_{1i} - d_{2i}$ for power cost laws. Equal variances and correlation one follow only if that difference is constant and slope variance is nonzero, whether or not equipartition holds. They are not a general test of ensemble equipartition.

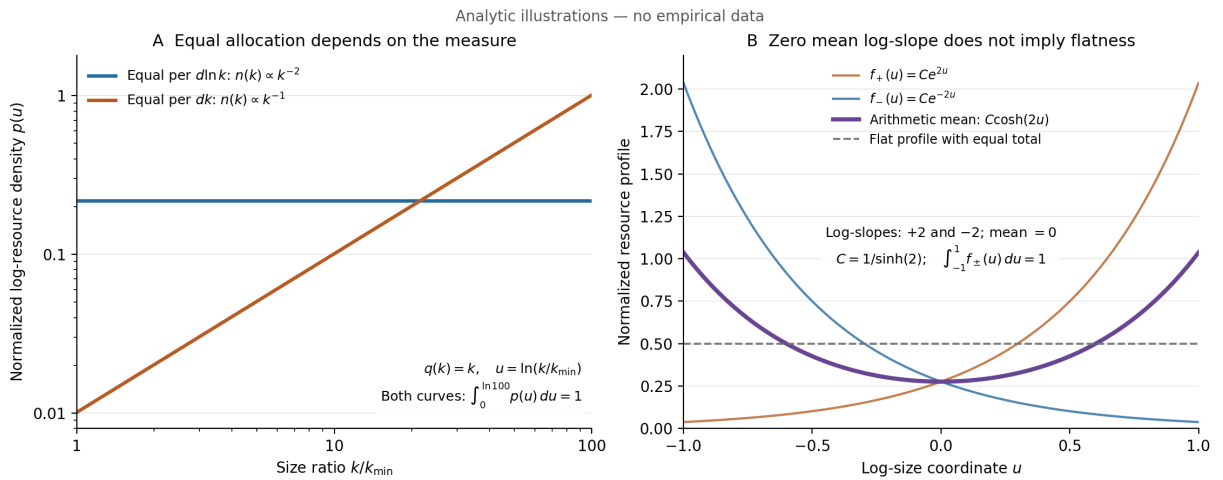


Figure 1. Analytic illustrations, not empirical data. (A) With resource equal to size, linear and logarithmic resource equipartition produce different count exponents. Both curves show unit-total resource density in the logarithmic coordinate. (B) Two unit-total resource profiles on a common domain have opposite log-slopes and therefore zero mean log-slope, but their arithmetic mean is not flat. Generated by run_theory.py; a vector version is available.

2.7 System constraints and observable distributions

Three levels of description must be kept distinct: an underlying resource-equalizing tendency, the neutral signature (2), and the outcome of the complete system. The latter depends on the proposed tendency together with resource costs, other processes, forcing, and initial and boundary conditions. A model containing an equalizing contribution can therefore predict a non-neutral stationary state or a transient with no clear neutral signature. Establishing such a contribution requires an explicit model or discriminating observations; equation (1) alone does not identify it.

Age and height distributions are motivating examples of constrained outcomes. A worked age-distribution account must specify birth history, survival, and migration; a height account must specify developmental processes, biological variation, and environmental conditions, as well as the population sampled. An orthopolity application must additionally identify the resource, its concentrations, and the comparison measure. No completed orthopolity model of either example

is supplied here. The relevant task is to derive the combined outcome and explain how its form depends on the constraints.

A failure of the neutral prediction is a finding about that prediction in the stated setting. It does not automatically establish the absence of the broader tendency, and counting distributions without its visible signature does not establish a general absence of evidence in Nature. A constrained model has its own predictions, which can succeed or fail. Specify the constraint through physical reasoning or independent measurements, derive its effect on the profile, and test the predicted response when it changes. Where recovery of neutrality is predicted, specify its conditions and timescale. This permits both neutral and non-neutral outcomes to inform the theory, while preserving the distinction between an observed departure, a proposed cause, and a tested explanation.

2.8 A mechanism on resource-concentration classes

Consider K fixed size classes, with positive reference widths w_i , fixed per-object resource costs q_i , nonnegative class stocks R_i , and conserved total $B = \sum_i R_i$. The classes represent populations of resource-bearing objects; the state is their aggregate resource. Equivalent counts $N_i = R_i/q_i$ may be continuous or expected counts, and their total need not be conserved. This effective model does not specify individual-object assembly or demographic kinetics. The full derivation states its interpretation and assumptions.

Let $g_{ij} = g_{ji} \geq 0$ be conductances on a connected undirected graph. Neutral exchange has current $J_{ij}^0 = g_{ij}(R_i/w_i - R_j/w_j)$ and dynamics $\dot{R}_i = -\sum_j J_{ij}^0$. Positivity follows from nonnegative jump rates g_{ij}/w_i ; antisymmetry of currents conserves total resource. Connectedness makes the stationary resource density constant, giving

$$R_i^0 = \frac{Bw_i}{\sum_j w_j}, \quad N_i^0 = \frac{Bw_i}{q_i \sum_j w_j}.$$

This is a mechanism selecting equal resource per declared class measure. The local exchange symmetry and the measure are assumptions of the model; they are not supplied by conservation alone. To model a constraint, prescribe dimensionless class preferences V_i and add downhill transitions:

$$a_{ij} = \frac{g_{ij}}{w_i} \exp(\max\{V_i - V_j, 0\}).$$

The rate a_{ij} is from i to j . Its neutral part g_{ij}/w_i remains, and the additional part is nonnegative. With column generator $A_{ji} = a_{ij}$ for $j \neq i$, $A_{ii} = -\sum_{j \neq i} a_{ij}$, define

$$\pi_i = \frac{w_i e^{-V_i}}{Z}, \quad Z = \sum_j w_j e^{-V_j}.$$

Then $\pi_i a_{ij} = \pi_j a_{ji}$ on every edge. Detailed balance selects the unique stationary resource $R_i^V = B\pi_i$. This uses standard reversible-chain mathematics (Levin and Peres, 2017), specialized here to fixed concentration classes and an explicit additive constraint mechanism. At that stationary state the neutral edge current is

$$J_{ij}^0 = \frac{Bg_{ij}}{Z}(e^{-V_i} - e^{-V_j}), \quad J_{ij}^c = -J_{ij}^0.$$

Thus a nonuniform stationary distribution can have an active equalizing contribution balanced by the constraint. Setting V to a constant removes the added rates and lets the actual constrained allocation relax toward neutrality.

For $p = R/B$, set $E^2 = \sum_i (p_i - \pi_i)^2 / \pi_i$. Writing $z_i = p_i / \pi_i$ gives

$$\frac{dE^2}{dt} = -2 \sum_{i < j} \pi_i a_{ij} (z_i - z_j)^2 \leq -2\lambda E^2,$$

where $\lambda > 0$ is the smallest nonzero eigenvalue of the symmetric matrix $-D_\pi^{-1/2} A D_\pi^{1/2}$. Consequently $E(t) \leq E(0)e^{-\lambda t}$ and $\text{TV}(p, \pi) \leq E/2$. The same bound applies after release, using the neutral generator and the state actually present at the switch. If connections are removed, each component retains its own resource total and only component-wise equalization follows.

The simulation fixes 24 equal-log size classes on $[1, 64]$, $q_i = k_i^{1.5}$, and adjacent conductances $1/\Delta \ln k$. A prescribed quadratic preference in $\ln k$, centered at $k = 8$ with width 0.75, produces a curved resource profile. The corresponding continuous count form is $dN/dk \propto k^{-2.5} e^{-V(\ln k)}$, a truncated lognormal shape for this quadratic choice. Each phase runs for eight relaxation times. The neutral and constrained spectral gaps are 0.569805 and 1.991712 per model time unit. The final relative errors in the neutral, constrained, and released phases are 4.73×10^{-4} , 2.41×10^{-7} , and 5.51×10^{-11} , each below its predicted bound. The opposing stationary drift vectors cancel to Euclidean residual 3.22×10^{-15} .

There are 4,096 equal-resource packets in each of 128 realizations, initially all in the first class. Exact matrix-exponential checkpoint transitions preserve the state at both switches. Packets remain independent, giving class-count law $X(t) \sim \text{Multinomial}(4096, p(t))$ and share covariance $(\text{diag}(p) - pp^T)/4096$. Uneven finite snapshots are therefore predicted even at neutral mean allocation. Rates, costs, preferences, and predictions were fixed before drawing the stochastic trajectories. The computational results and checks are retained with the model report; the simulation supplies a constructive example, not new natural-system observations.

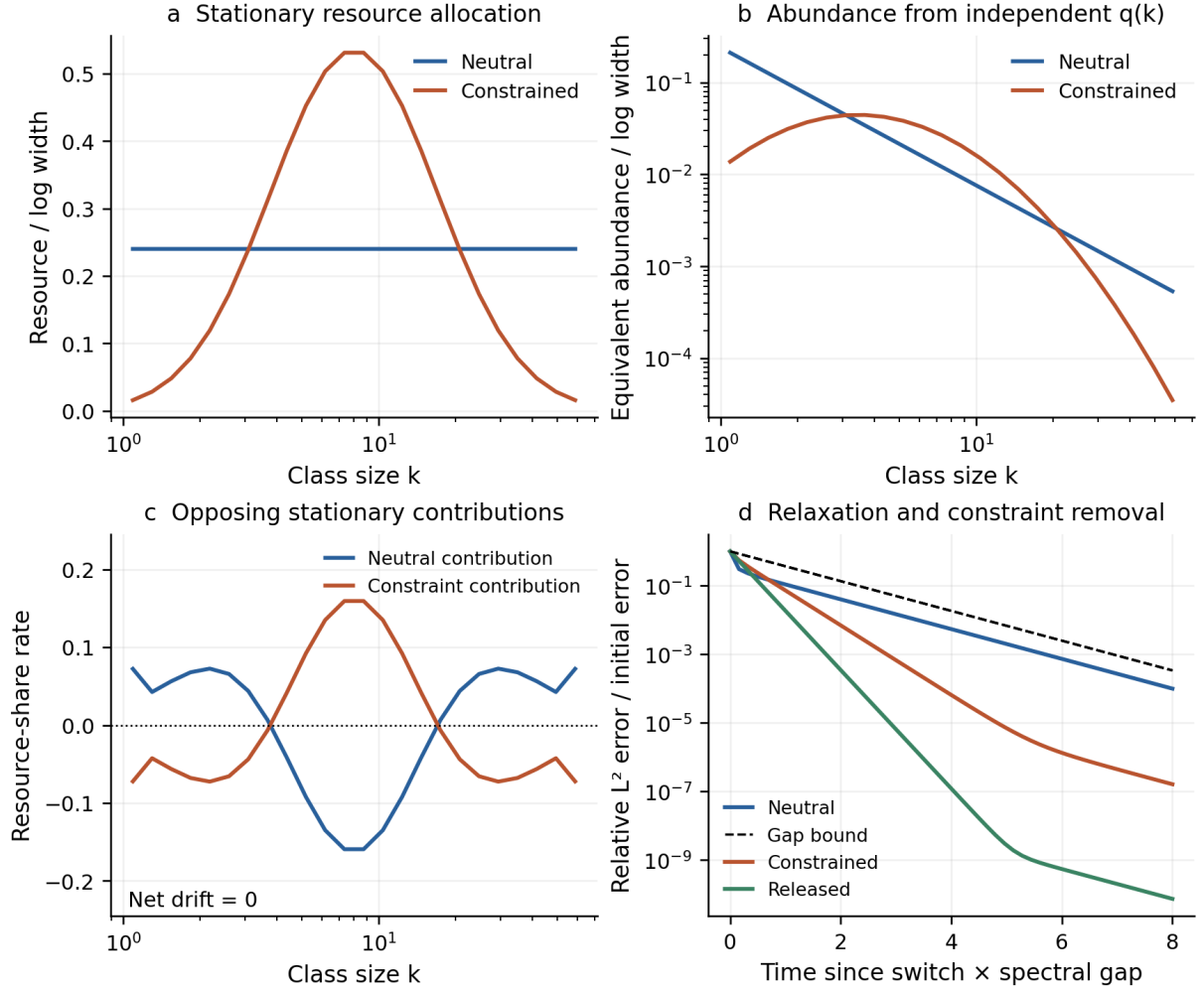


Figure 2. Fixed class-exchange model. Equalizing kinetics select neutral resource shares; an added class preference selects a curved profile and changes equivalent abundance. At constrained equilibrium the neutral and added contributions cancel. Removing the preference preserves the current state and predicts recovery under the neutral dynamics. The displayed convergence curves are deterministic expectations; stochastic packet realizations and their uncertainty are retained in the linked report.

2.9 A physical realization from electromagnetic transport

For electromagnetic energy density u and Poynting vector $\mathbf{S}$, Maxwell's equations give $\partial_t u + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}$ (Poynting, 1884; Haus and Melcher, 1989). Define $P(r) = \int_{S_r} \mathbf{S} \cdot \mathbf{n} dA$ outside a source enclosed by an inner sphere r_0 with steady power $P(r_0) = L$. Integrating the balance over a transparent, source-free shell with no net energy accumulation gives $P(r_2) = P(r_1) = L$. Isotropy is a separate condition: it makes the local radial flux density $F = L/(4\pi r^2)$; otherwise this is its sphere average.

Under outgoing radial transport at constant speed c , without scattering or secondary emission in the modeled component, a specified absorber with coefficient $\kappa(r)$ gives

$$\frac{dP}{dr} = -\kappa(r)P, \quad P(r) = L \exp \left[- \int_{r_0}^r \kappa(s) ds \right].$$

This is the absorption law specialized to spherical transport (Condon and Ransom, 2016). An absorbing layer with optical depth $\ln 2$ halves downstream power. The missing radiative power is deposited in matter, so the modeled radiation component has an open budget. Its departure follows from the prescribed medium and the same energy balance.

These nested surfaces receive the passage of the same energy; their powers cannot be summed as disjoint resource shares. Disjoint shell stocks do give an additive comparison: in transparent stationary radial transport, $dU = (L/c)dr$. Thus equal radial thicknesses hold equal energy, whereas $dU/d \ln r = Lr/c$ increases with radius. The resource, units, and comparison measure determine which equality is physically meaningful.

The response to removing an absorber is also determined. For sphere-averaged u , with $e = 4\pi r^2 u = P/c$, the transport equation is $\partial_t e + c \partial_r e = -c\kappa(r, t)e$. Suppose the source remains constant, the initial field is stationary under $\kappa_{\text{old}}(r)$, and absorption is set to zero throughout the modeled domain at $t = 0$. For $t \geq 0$, characteristics give

$$P(r, t) = L \exp \left[- \int_{r_0}^{\max\{r_0, r-ct\}} \kappa_{\text{old}}(s) ds \right].$$

Every radius recovers by $(r - r_0)/c$. Downstream of a layer $[a, b]$, the response starts at $(r - b)/c$ and completes at $(r - a)/c$. This is an idealized change of the absorption coefficient, with no radiation injected by the removal process. The physical derivation states the boundary history and all transport assumptions. It supplies an established physical realization of a declared equality and its constrained departure; it does not equate radial shells with organism size classes or derive abundance laws from an inverse-square flux.

2.10 Classical equipartition and quantum radiation

Thermal electromagnetic radiation supplies a second physical realization with a different reference measure: counting field modes, including polarization. For a finite set of classical normal modes in canonical equilibrium at $T > 0$, each has two quadratic terms, $H_j = P_j^2/(2m_j) + m_j \omega_j^2 Q_j^2/2$. Gaussian integration over both coordinates gives $Z_j^{\text{cl}} \propto (\beta \omega_j)^{-1}$, with $\beta = (k_B T)^{-1}$, and therefore

$$\langle H_j \rangle = -\partial_\beta \ln Z_j^{\text{cl}} = k_B T.$$

A class of M_i modes consequently has mean energy $M_i k_B T$: the resource density is equal per mode, with independently defined weight $w_i = M_i$. This is equality of equilibrium means, not identical instantaneous energies. Equilibrium preparation or thermal coupling must be justified; uncoupled harmonic modes do not thermalize each other merely because their canonical

distribution can be written down. This argument uses standard equipartition and supplies no relaxation time (Tong, n.d.).

Quantization changes the allowed states. Define the resource as thermal excitation energy above the ground state, excluding the temperature-independent zero-point term. For photon absorption and emission in thermal equilibrium, photon chemical potential is zero and a mode has excitation levels $\epsilon_n = nh\nu$. The geometric sum gives

$$Z_\nu = \frac{1}{1 - e^{-x}}, \quad \bar{\epsilon}_\nu = \frac{h\nu}{e^x - 1} = k_B T f(x), \quad x = \frac{h\nu}{k_B T}, \quad f(x) = \frac{x}{e^x - 1}.$$

For $x > 0$, f decreases strictly from 1 toward 0. Its limits are $f(x) = 1 - x/2 + x^2/12 + O(x^4)$ near zero and $f(x) \sim xe^{-x}$ at large x . The quantum level spacing relative to thermal energy specifies the departure before any spectrum is observed. Raising temperature or lowering frequency recovers classical relative equality. This is a restriction of the available energy states; it does not establish the additive decomposition into opposed classical currents proved in Section 2.8.

In a large three-dimensional vacuum cavity, counting two polarizations gives the mode density $dM/(Vd\nu) = 8\pi\nu^2/c^3$. Multiplication by $\bar{\epsilon}_\nu$ yields the Planck energy density and, for isotropic radiation, the spectral radiance:

$$u_\nu(T) = \frac{8\pi h\nu^3}{c^3(e^{h\nu/(k_B T)} - 1)}, \quad B_\nu(T) = \frac{c}{4\pi} u_\nu(T).$$

Thus the classical equality per mode gives $u_\nu^{\text{cl}} \propto \nu^2$ and $dU/(Vd\ln\nu) \propto \nu^3$, neither a flat linear-frequency nor a flat log-frequency spectrum. If photons are the counted objects, $q(\nu) = h\nu$ and their mean number per mode approaches $k_B T/(h\nu)$ in the classical regime. Extending classical equipartition to infinitely many modes gives divergent total energy; quantization makes it finite, $U/V = 8\pi^5 k_B^4 T^4/(15h^3 c^3)$. These are established results (Planck, 1901; Tong, n.d.; Condon and Ransom, 2016), here specifying the resource, comparison measure, and constraint in the allocation framework.

2.11 Connection to the measured thermal spectrum

FIRAS measured the cosmic microwave background against calibrated blackbody references and separated monopole, dipole, and Galactic components (Fixsen et al., 1996). NASA LAMBDA's distributed monopole table contains a 2.725 K blackbody plus the residuals from that published analysis, with marginal uncertainties and a Galactic model column. The residuals already follow temperature and foreground fitting. The table therefore supports a transparent re-expression of the published result; agreement with its embedded template cannot be counted as a new independent confirmation.

Our retrospective calculation retains all 43 rows and fixes $T_0 = 2.725$ K from the product definition. With wavenumber $\tilde{\nu}$ in inverse centimetres, $\nu = 100c\tilde{\nu}$. Radiance remains per hertz, with $1 \text{ MJy/sr} = 10^{-20} \text{ W m}^{-2} \text{ Hz}^{-1} \text{ sr}^{-1}$. Dividing the inferred energy density by the independent mode density gives

$$\frac{\hat{\epsilon}_\nu}{k_B T_0} = \frac{c^2 I_\nu}{2\nu^2 k_B T_0} = \frac{I_\nu}{B_\nu^{\text{RJ}}(T_0)}, \quad B_\nu^{\text{RJ}} = \frac{2\nu^2 k_B T_0}{c^2}.$$

The channels span 68.05–639.46 GHz and $x = 1.199$ –11.262. Their transformed mode energies range from 0.518 to 1.32×10^{-4} of $k_B T_0$ (Figure 3). They illustrate the quantum departure, not the $x \ll 1$ classical plateau. The CMB can retain a Planck spectrum through collisionless cosmological redshifting; this example does not require current thermal contact with surrounding matter (Condon and Ransom, 2016, Section 2.6.2).

For residual diagnostics we use the source’s original residual column, not the difference between rounded monopole intensities and a newly calculated curve. The source’s approximate frequency covariance is $C_{ij} = \sigma_i \sigma_j Q_{|i-j|}$, using its 43 published lag coefficients (Fixsen et al., 1996, Section 3.3). The matrix is positive definite. Applying the linear radiance-to-mode conversion to both axes propagates this covariance to the mode-energy ratios. The residual quadratic form is $r^T C^{-1} r = 49.774$; it is a descriptive audit of the published approximation, with no new fit or goodness-of-fit probability. It need not reproduce the original full-analysis statistic, 46 for 40 degrees of freedom. Diagonal inverse-variance weighting gives residual RMS 18.90 kJy/sr, or 49.28 parts per million of the tabulated peak radiance. This defined amplitude summary does not assume that the channels are independent.

Using exact SI constants at the rounded source coordinates gives a discrepancy $I_{\text{posted}} - [B_\nu(T_0) + r_{\text{published}}]$ of -0.127 to $+7.598$ kJy/sr, with radiances in common units. This discrepancy is retained without assigning a cause or replacing the original residuals. Absolute calibration, foreground modeling, and the prior fits remain inherited from the published analysis. The thermal-radiation report retains the full derivation, source bytes, covariance transcription, and reproducible calculation. This example connects a physical equality and an independently derived quantum departure to established observational evidence under its documented conditions.

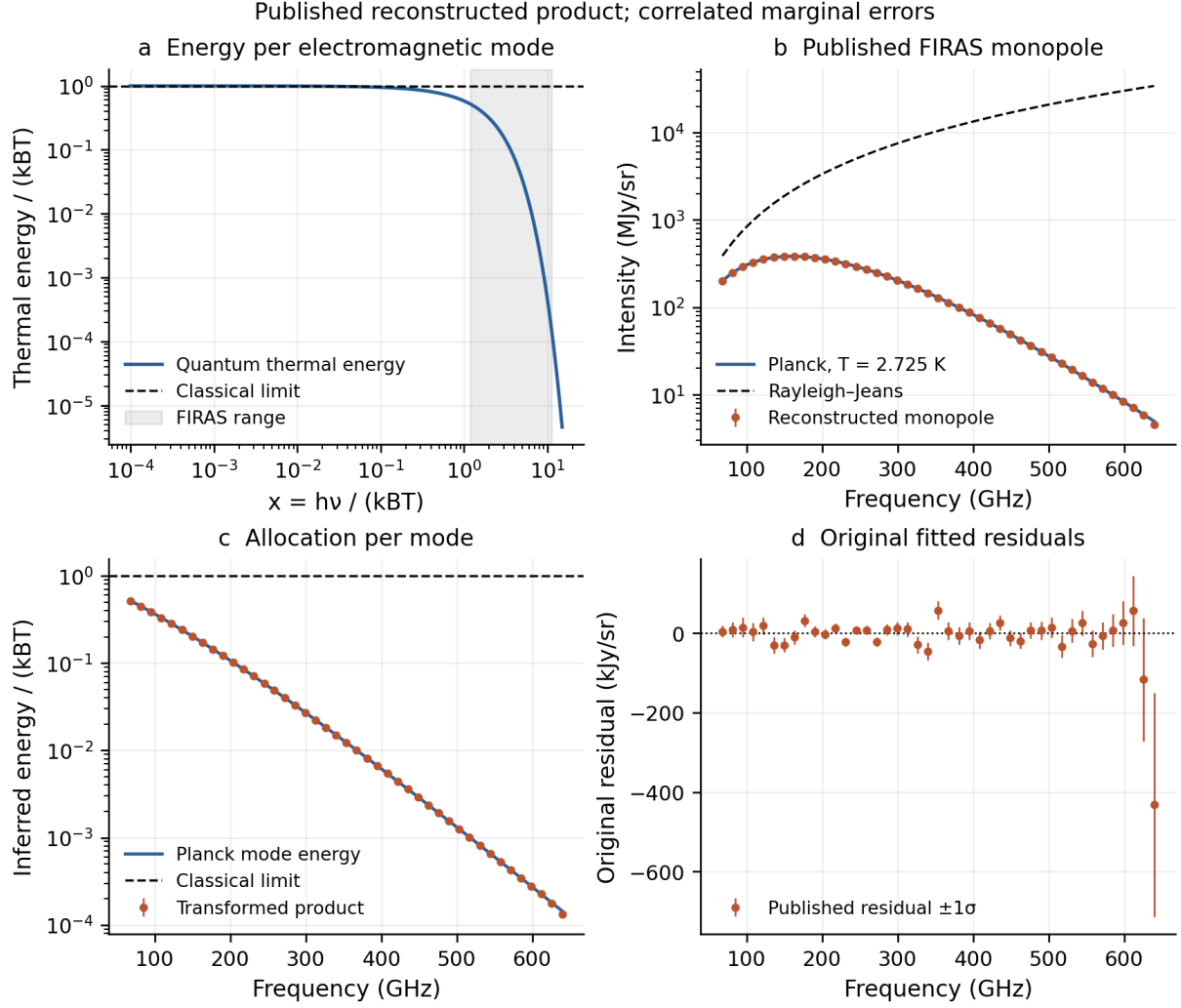


Figure 3. Thermal energy per electromagnetic mode. (A) The theoretical quantum factor $f(x)$ approaches classical equality as $x \rightarrow 0$. (B) The NASA FIRAS reconstructed monopole product and Planck and Rayleigh–Jeans radiances at the product’s 2.725 K reference temperature. (C) The same product expressed as energy per mode relative to $k_B T_0$. (D) Original published residuals with marginal one-sigma uncertainties; channels are correlated. The Planck template is part of the distributed product, so panels B and C illustrate the published result rather than provide a new independent model test. All 43 channels are retained.

3. Data and methods

3.1 Sources and scope

Data source	Analysed material	Resource and size coordinate
NOAA GOES XRS flare reports, 2022–2024	10,501 catalogue events; 1,168 paired events in the primary evaluation domain	End fluence in J/m ² ; peak irradiance in W/m ²
USGS ComCat, 2010–2024	6,639 events after retaining moment-magnitude types and rounding to 0.1 at threshold 5.5	Magnitude-derived energy proxy as both resource and coordinate
Hatton et al. (2021)	23 one-decade biomass bins, upper 200 m and full water column	Body mass for both
GLOSSAQUA (Ersoy et al., 2025)	1,300 normalized biomass-spectrum estimates from 16 study identifiers	Body mass for both

These are four data sources; the two ocean domains are related reconstructions. GLOSSAQUA contains published estimates rather than organism-level observations. Study identifiers do not guarantee independent authors, methods, sites, or measurement errors. The sources are convenience cases for evaluating a hypothesis, not a representative sample of natural systems. Input bytes are fixed by SHA-256 manifests; provenance and source notices accompany the data.

3.2 Analysis-plan provenance

The pilot configuration explicitly labels itself exploratory. The subsequent aquatic/ocean plan is retained as prereg_2026-09-09.json. In local history, commit 6df364f contains the plan and raw slope data, preceding analysis commit a0c9973. This records an order of commits. It does not establish an immutable public registration, independently verified blinding, or when values were first viewed. Statements of blinding in the historical file are author reports. The current corrections are retrospective, and the later ensemble, stratification, and variance analyses are exploratory. No claim of confirmatory preregistration is needed for the descriptive and mathematical results here.

3.3 Accounting and uncertainty

For bins B_j of logarithmic width w_j , compute $\hat{O}_j = \sum_{i \in B_j} q_i / w_j$ and $\hat{C} = \sum_j \sum_{i \in B_j} q_i / \sum_j w_j$. The normalized profile $\phi_j = \hat{O}_j / \hat{C}$ has width-weighted mean one by construction. This is not evidence of flatness. Empty bins remain in the profile, the final boundary is included, and missing resources require an explicit exclusion policy.

A factor- F endpoint drift across log-domain width L corresponds, for a power-shaped spectrum, to slope tolerance $c = \ln(F)/L$. This differs from requiring every $\phi_j \in [1/F, F]$: the latter permits a maximum-to-minimum ratio of F^2 . Both slope and profile deviations matter. A slope equivalence interval combined with a point-estimate profile screen is not full simultaneous equivalence; that would require uncertainty for the whole profile. Failure to establish equivalence is not, by itself, evidence of a nonzero departure.

For flares, the arithmetic mean resource exponent is fitted by Gamma quasi-likelihood on 2022 data, with bounded count-density fitting on 2023–2024 over $[10^{-5}, 10^{-3}]$ W/m². Log-resource least squares would target a different conditional quantity unless an appropriate retransformation were justified. Pilot intervals use 600 calendar-month block resamples for flares and calendar-year

blocks for earthquakes. These do not capture all dependence or selection bias. The temporal split separates estimation samples but does not make the exploratory domain choice a prospective prediction.

Power-law diagnostics use maximum likelihood and a refitted parametric-bootstrap KS statistic on declared support, following the approach of Clauset et al. (2009). The rounded earthquake magnitudes are tested on their integer grid: for $K = (M - M_0)/\Delta$, $P(K = k) = (1 - r)r^k$ with $r = 10^{-b\Delta}$. Continuous-reference KS calibration is unsuitable for these ties. Parametric-bootstrap probabilities remain conditional on an independent-event working model; catalogue clustering is a limitation.

3.4 Aquatic estimates and reconstruction uncertainty

The primary GLOSSAQUA subset requires body mass on the size axis, finite slopes and positive ordered size limits, and the normalized biomass-spectrum convention. For biomass per unit linear mass $B(m) \propto m^t$, equal biomass per log mass requires $t = -1$. We examine $\beta = t + 1$ and each estimate's reported size span. The historical plan uses $F = 1.25$ and $F = 2$, a 2,000-replicate study-block bootstrap for the pooled median, and a joint criterion involving its interval and the fraction of slopes inside their tolerances. That fraction ignores individual slope uncertainty and must not be interpreted as formal individual equivalence. Slope summaries cannot detect within-spectrum curvature. A post-audit sensitivity explicitly excludes StudyID_07 from range-dependent assessment because its reported bounds are physically implausible; no replacement bounds are imputed (§4.4).

For the ocean reconstruction, 20,000 simulated profiles propagate reported multiplicative uncertainty factors using lognormal draws centred on the published log estimates, with log-scale standard deviation $\ln(f)/1.96$. The published bounds match estimate divided and multiplied by f . Independent errors across groups and bins are a working assumption, not a property established by those bounds. The reconstruction is not a sample of independent organisms, and its plateau was selected after inspecting the upper-ocean estimates.

Exploratory random-effects and latent-distribution fits use reported errors where available. Intervals converted to standard errors assume their advertised coverage and approximate normality. Size-spectrum estimators can have biased estimates and miscalibrated intervals (Edwards et al., 2017, 2020). Repeated spectra within studies and estimated error variances further limit inference; independence-based pooled intervals are not primary evidence of a population mean.

4. Results

4.1 Earthquakes contradict the specified energy-proxy allocation

Gutenberg–Richter scaling $N(\geq M) \propto 10^{-bM}$ and proxy $E \propto 10^{\gamma M}$ give resource per log energy proportional to $E^{1-b/\gamma}$. Flatness therefore requires $b = \gamma$. At threshold 5.5, the fitted $b = 0.998$ has year-block 95% interval $[0.973, 1.024]$, far below the chosen $\gamma = 1.5$. Thresholds 6.0 and 6.5 give 0.984 and 0.977. Using conversion exponent 1.44 also leaves a substantial discrepancy.

At threshold 5.5 the discrete KS statistic is approximately 0.0077, with a parametric-bootstrap probability about 0.26. This is non-rejection of the fitted geometric model, not proof that the model is exact. A power-like tail can thus coexist with a resource-proxy allocation that disagrees with $O_{\log}$. No discrete lognormal comparison was conducted. Magnitude-derived energy is not independently measured radiated energy; the finding is restricted to the specified proxy and catalogue, without declustering or a regional completeness model.

4.2 Flares disagree with the joint prediction on the selected domain

The fitted resource exponent is $d = 0.858$ with 95% interval $[0.697, 1.054]$. The implied density exponent is 1.858, compared with 2.239 $[2.085, 2.399]$ in the paired evaluation sample. The estimated gap is 0.382 $[0.125, 0.620]$. A rise-phase fluence sensitivity gives gap 0.411 $[0.208, 0.604]$. This alternate resource has no missing values in the domain, but does not prove the primary end-fluence analysis free of missingness bias.

For all 1,306 in-domain evaluation events, peak irradiance has fitted exponent 2.274 and KS statistic about 0.032; its bootstrap probability is about 0.02. The power law is therefore inadequate under this working test. Its likelihood cannot clearly distinguish it from the fitted lognormal. Exponent mismatch tests the conjunction of equipartition, power-shaped mean cost, transfer between years, and the count model; it cannot isolate equipartition from all those assumptions.

The gaps at four lower cutoffs are -0.480 , 0.191 , 0.382 , and 0.960 . This substantial sensitivity discourages treating one selected range as a universal scaling regime. Fluence is instrument-band irradiance integrated over an event, not total flare energy.

4.3 The ocean reconstruction is compatible with a broad, uneven plateau

The upper-ocean summary reproduces the published abundance slope at approximately -1.039 . Its maximum-to-minimum biomass ratio is 38.8 across all 23 bins, and 1.7 within the selected plateau of 16 bins. The corresponding full-water-column plateau ratio is 4.3. The 23 one-decade bins span 23 decades between outer edges; their centres are 22 decades apart. Similarly, the plateau covers 16 decades between edges and 15 between centres.

Under the stated uncertainty propagation, only 2 of 23 upper-ocean bin intervals lie wholly outside the factor-1.25 band; the corresponding full-column count is 3. These are pointwise intervals without multiplicity adjustment. The remaining intervals can include both near-flat and materially non-flat values, so their overlap with the band establishes neither equality nor absence of departures. Correlated reconstruction errors and post hoc range selection further limit the assessment. These results re-express Hatton et al.’s reconstruction; they do not independently replicate it.

The full upper-ocean range has a fitted biomass log-slope of -0.0392 and a propagated 90% interval approximately $[-0.058, -0.026]$, outside the factor-1.25 slope tolerance of 0.00421. Thus the assumed uncertainty model indicates a systematic gradient even though most individual bin intervals overlap the pointwise band. Pointwise overlap does not make the whole-profile question uninformative; its interpretation remains conditional on the reconstruction error model.

4.4 Aquatic slopes are close in location, without established equivalence

Among 1,300 normalized biomass spectra from 16 study identifiers, the pooled median slope is -1.015 . The study-block 95% interval for its departure from -1 is $[-0.100, 0.010]$. The median slope tolerance is 0.0317 at $F = 1.25$ and 0.0984 at $F = 2$. This interval is not wholly within either band. Under the historical joint criteria, the result is **not supported**.

The reported-slope fractions inside their individual drift tolerances are 0.07846 ($F = 1.25$) and 0.24231 ($F = 2$), counting boundary cases inclusively. These are descriptive compatibility fractions. Their complements are not estimates of the fraction of true spectra violating equipartition, because measurement error and within-spectrum shape are not accounted for.

These historical fractions also depend on a metadata defect: all 377 StudyID_07 records report bounds of 2×10^{-8} to 2×10^{27} pg C, a 35-decade range with a physically implausible upper body

mass of 2×10^{12} kg C. Excluding those records from this range-dependent calculation leaves 923 slopes from 15 studies. Their point-slope compatibility fractions are 9.32% at $F = 1.25$ and 28.93% at $F = 2$. The source-derived replacement bounds are unknown; the original slopes remain in the location summary. The remaining records also require a methods and units audit. Consequently neither version of the pass fraction should be treated as a validated estimate of ecological prevalence.

Two studies supply 1,016 of the 1,300 estimates, approximately 78%. A central estimate near -1 is worth investigating, but it neither establishes a mean of -1 in a defined population nor establishes equal average biomass occupancy. The stronger ensemble interpretation fails on both statistical and mathematical grounds (§2.4).

4.5 Exploratory diagnostics identify further limits

Latent-shape fits and random-effects summaries describe appreciable variation in the error-reporting subset. Their fitted dispersion includes ecological differences, study methods, and potentially unmodelled dependence. An I^2 estimate is not a measured fraction of variance caused by ecosystem differences. AIC rankings among Gaussian, Laplace, and Student families do not establish adequacy or identify ecological classes. Error-aware, matched-sample pass fractions are internal model checks, not independent predictions of individual failure.

The cross-study comparison of 639 lake-fish estimates with 377 records associated with Gaedke’s Lake Constance study does not partition spatial and temporal variance. Different taxa, sampling designs, fitting methods, and measurement uncertainties prevent treating their variance ratio as a bound on one population. Reproducing Arranz et al.’s reported summary statistics from the same underlying observations checks extraction, not independent replication of a dispersion parameter.

Exponent metadata also need source verification. Counts in equal-log bins and abundance divided by linear bin width have slopes differing by one under power scaling. The Perkins et al. (2018) count-bin method illustrates why a database category alone may not identify the estimand. Choosing a mapping because it makes a slope closer to -1 or -2 would bias the test. Secondary conventions remain sensitivity analyses; a complete study-level methods audit is needed before pooling them. Checks of two dominant primary studies do not validate every record in the primary subset.

5. Frozen-forecast tests

The analyses of Section 4 assess data that had already been inspected. We therefore added tests in which the resource, its cost law, the forecasts and the decision rules are fixed before the evaluated outcomes are decoded, and in which whole units are held out: launches, a calendar year, vessels and selection treatments. Each executed study retains its inputs, frozen protocol, archived algorithms, forecasts and results in an append-only run registry. Three kinds of evidence are kept separate: measurements on engineered systems, an unused observation period of a natural catalogue, and published biological experiments reanalysed retrospectively. Published summaries of those experiments had been read, so their reanalyses are retrospective validations, not blinded tests. A further plant study uses fixed scoring and a development-only geographic split after accidental raw outcome exposure. Its freeze cannot establish that those outcomes were unseen.

5.1 Controlled allocation measurements

In a controlled workload experiment, separately measured memory and CPU costs of matrix tasks predicted completion profiles under assigned quota pairs within four tolerances frozen before 672 validation tasks (workload pilot). A fresh launch exposed a transfer boundary: the original forecast failed its unchanged tolerance in two of four conditions, and local recalibration recovered one (workload transfer). The quotas were acceptance criteria, so these tests concern the transfer of cost models, not allocation.

An allocation test requires the system, not the experimenter, to divide a resource. In a Python runtime whose global interpreter lock serializes execution, worker threads running matrix kernels of five sizes competed for one execution resource (dimensionality intervention). Separate calibration gave CPU cost degrees 1.870 and 2.686 for the two kernels. The calibrated mean-cost curves predicted complete CPU and job profiles before 16 allocation trials, with maximum CPU-share errors of 0.0035–0.0063 and job-share total-variation errors of 0.0021–0.0124. Replacing the largest-class thread with a second smallest-class thread produced the predicted CPU-share change within 0.0078. A unit-cost alternative had count errors of 0.126–0.273. This success concerns one engineered allocation mechanism; it identifies no geometric dimension and no tendency of natural systems.

5.2 Calibrated decisions and an unused observation period

Complete-profile equivalence decisions were calibrated on 180,000 independently generated datasets in 90 conditions (profile calibration). Simultaneous bands met the declared gate for dense, bounded, independent blocks at 48 blocks, with minimum coverage 93.95%, but failed at 12 and 24 blocks. With twelve serially dependent blocks, wrong-departure rates reached 43.75%. These failures define observation designs for which no neutrality verdict is available, whatever a nominal interval suggests.

For solar flares, measured rise-phase fluence costs, six logarithmic classes, three forecasts, a tolerance and an application gate were committed before the 2025 NOAA records were acquired (solar validation). Of 3,277 events, 356 met the domain rules. Logarithmic neutrality predicted the class counts much better than linear neutrality, with count total variation 0.125 against 0.573; historical counts did better still, at 0.022. The point profile departed from the $\ln 1.5$ margin, but the calibrated decision remained unresolved, as declared before acquisition, and the instrument mix changed between periods.

5.3 Published biological measurements

Study	Held out	Resource cost	and	Frozen comparison	Outcome
<i>Synechococcus</i> quotas (Harcourt et al., 2024)	Ten cultures at 25 °C	C, N and P per cell against measured diameter		Fixed cube, free power, strain means, temperature trends	Fixed cube best for C; strain means best for N and P
Size-selected <i>Dunaliella</i> (Malerba et al., 2018)	Each selection treatment	Biovolume at carrying capacity in one shared medium		Equal biovolume ($d = 1$), fitted size law, assigned exponents, equal cell number	Equal biovolume best; cells at capacity scale as $V^{-1.02}$
Food-web chemostats (Wojcik et al., 2025)	Twelve polyculture vessels	Algal N stock: biovolume times separately assayed N per volume		Persistence, development response, no-herbivore response, equal stock, two-budget cost ratio	No model beat persistence; the cost-ratio rule failed
Harvested plants (Dillon et al., 2019)	Five Colorado plots; prior raw exposure	Direct above-ground dry mass; $q(m) = m$ by definition		Log and linear neutrality, Ohio histogram, bounded Pareto and Weibull	Pareto best for biomass; trained models better for counts; ecological neutrality unresolved

Published elemental quotas of four *Synechococcus* strains tested whether a size-based cost transfers to an unused temperature (cost transfer). Trained on 16–22 °C cultures, a cubic diameter law with one fitted intercept predicted held-out 25 °C carbon quotas within a typical factor of 1.14. For nitrogen and phosphorus, strain means beat both size laws, and a fitted exponent never beat the fixed cube. Extrapolated temperature trends were worst for every element. A cost calibration can therefore transfer for one resource while failing as a size law for another resource in the same cells.

The accounting identity behind orthopolity, $Nq(V) = R$, predicts how abundance scales with per-object cost when a budget is shared. Thirty *Dunaliella tertiolecta* lineages, artificially selected for about 280 generations into small, control and large classes, were regrown separately in one F/2 medium after replete, N-deprived or P-deprived histories (size budget). The protocol was committed before any small- or large-selected outcome was decoded; each held-out selection treatment was then predicted from the other two. Across a 10.4-fold range of mean cell volume, replete carrying capacity in total biovolume was constant to within 3%, so cell number at capacity scaled as $V^{-1.02}$ (Figure 4). The frozen equal-biovolume law ($d = 1$) had held-out errors of 0.124 and 0.179 log units. It outperformed a size law fitted within part of the range, whose fitted cost dimension was 0.74 in one fold and 1.12 in the other, and an equal-cell-number law, with errors of 1.48–2.08. A carbon cost degree assigned from the *Synechococcus* study ($d = 0.91$) performed comparably; its nitrogen degree ($d = 0.80$) did not. Regrowth after N deprivation nearly restored capacity, whereas P deprivation left an overshoot of 25–31% in control and large lineages but not in small ones. Restoring the medium therefore did not guarantee a return to the replete profile.

In 24 food-web chemostats, a nitrogen pulse redistributed algal resource composition (chemostat response). Pre-pulse N and C per cell volume from separate monocultures converted three algal groups’ biovolumes into resource stocks. Forecasts for twelve held-out polyculture vessels were frozen before their post-pulse values were decoded. The pulse moved resource composition by 0.25 total variation on average, about 1.5 times the pre-pulse variation around baseline, yet no

forecast improved appreciably on persistence: 0.243 for the development response against 0.247. A two-budget allocation rule, in which a binding secondary carbon budget reweights each group's share by its measured C:N ratio, failed on three counts. Its attainable redistribution, 0.05–0.12, was below every observed departure, 0.29–0.49. Its predicted direction held in 4 of 12 vessels, the chance count. It did not beat persistence. No held-out vessel recovered its pre-pulse composition within 12 days.

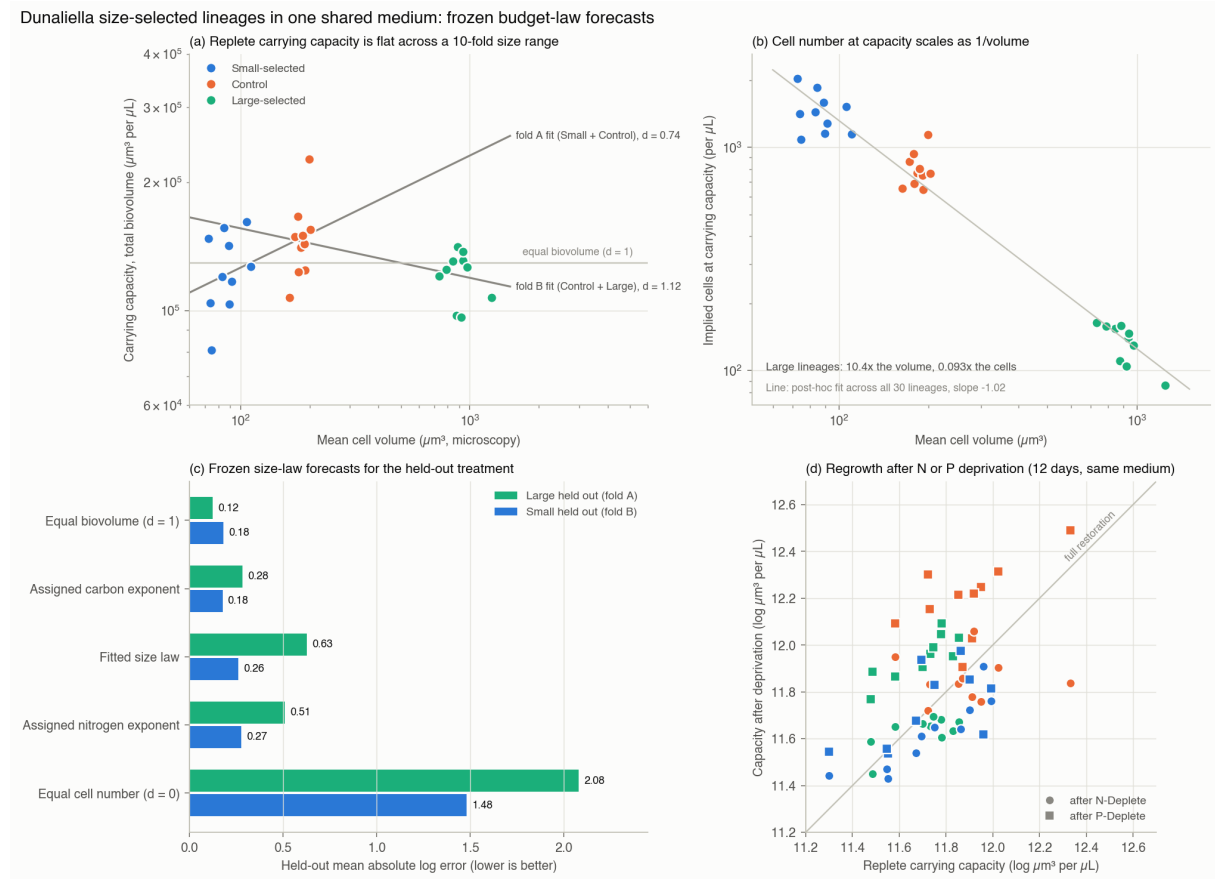


Figure 4. Size-selected *Dunaliella* lineages regrown in one shared medium, from the data of Malerba et al. (2018). (a) Replete carrying capacity in total biovolume against mean cell volume, with the two cross-fitted size laws and the equal-biovolume law. (b) Implied cell number at capacity. (c) Held-out errors of the frozen forecasts. (d) Capacity after N or P deprivation against replete capacity. Generated from retained outputs by `report_dunaliella_size_budget.py`.

5.4 Directly weighed coexisting plant stocks

Ten harvested herbaceous plots provide direct aboveground dry masses for coexisting ramets or stem clusters (Dillon et al., 2019; plant study). Forest and desert allometric masses are excluded. Five Ohio plots supply development fits; five Colorado plots supply evaluation. The protocol, algorithms and forecasts were committed before formal scoring, but raw rows had already been exposed during source inspection. This is a retrospective transfer comparison. It identifies neither a limiting nutrient nor an opportunity budget; $q(m) = m$ defines the measured stock rather than testing an independent cost law.

The frozen domain is 0.01–100 g, split into eight half-decade bins with empty classes retained. The upper bound depends only on Ohio masses. Thirty missing mass records, 144 subthreshold ramets and one above-domain ramet remain in the evaluation membership ledger. The latter weighs 112.55 g and accounts for 24.50% of one plot's known biomass. In-domain mass coverage

is 75.48% there and 99.89–99.97% in the other four plots. Scores therefore describe the declared domain; the excluded mass and unknown missing mass cannot be silently absorbed.

The primary score is equal-plot mean total variation between realized normalized biomass and each expected-stock or empirical template. Bounded Pareto scores 0.470, logarithmic neutrality 0.493, linear neutrality 0.507, the development histogram 0.510 and bounded Weibull 0.531 (Figure 5). Pareto beats logarithmic neutrality in three of five plots; large opposing plot differences leave a mean improvement of only 0.023. Count profiles provide a different comparison: the development histogram, Pareto and Weibull score 0.260–0.271, versus 0.412 for linear and 0.495 for logarithmic neutrality. Their binned count log losses are about 1.83, versus 2.08 and 2.42. These are point discrepancies, without a calibrated ecological superiority or equivalence verdict. They also do not contradict the original study’s superior within-site Weibull fits: the models here transfer across locales and share a development-fixed domain.

Synthetic calibration fixes census size and draws masses from m^{-2} on the same domain, so each bin has equal expected biomass before measurement rounding. At 160 independent ramets, mean realized stock TV is 0.441 and every simulated census has an empty bin. The largest bin’s average normalized share is 3.13%, although its normalized expected stock is 12.50%. At 1,280 ramets these values are 0.244, 87.2% and 9.12%. Thus $E[R_i / \sum_j R_j]$ differs from $E[R_i] / \sum_j E[R_j]$. A separately generated iid 95% TV envelope has exceedance rates 6.2% and 5.2%, but repeating masses in blocks of twenty raises these to 97.8% and 99.9%. This stress test is not an ecological dependence model. Field dependence, inclusion and independent regional replication remain unidentified, so the ecological neutrality decision is unresolved. An uneven small census alone cannot settle a claim about expected allocation.

Measured plant dry mass: retrospective profile transfer and census calibration

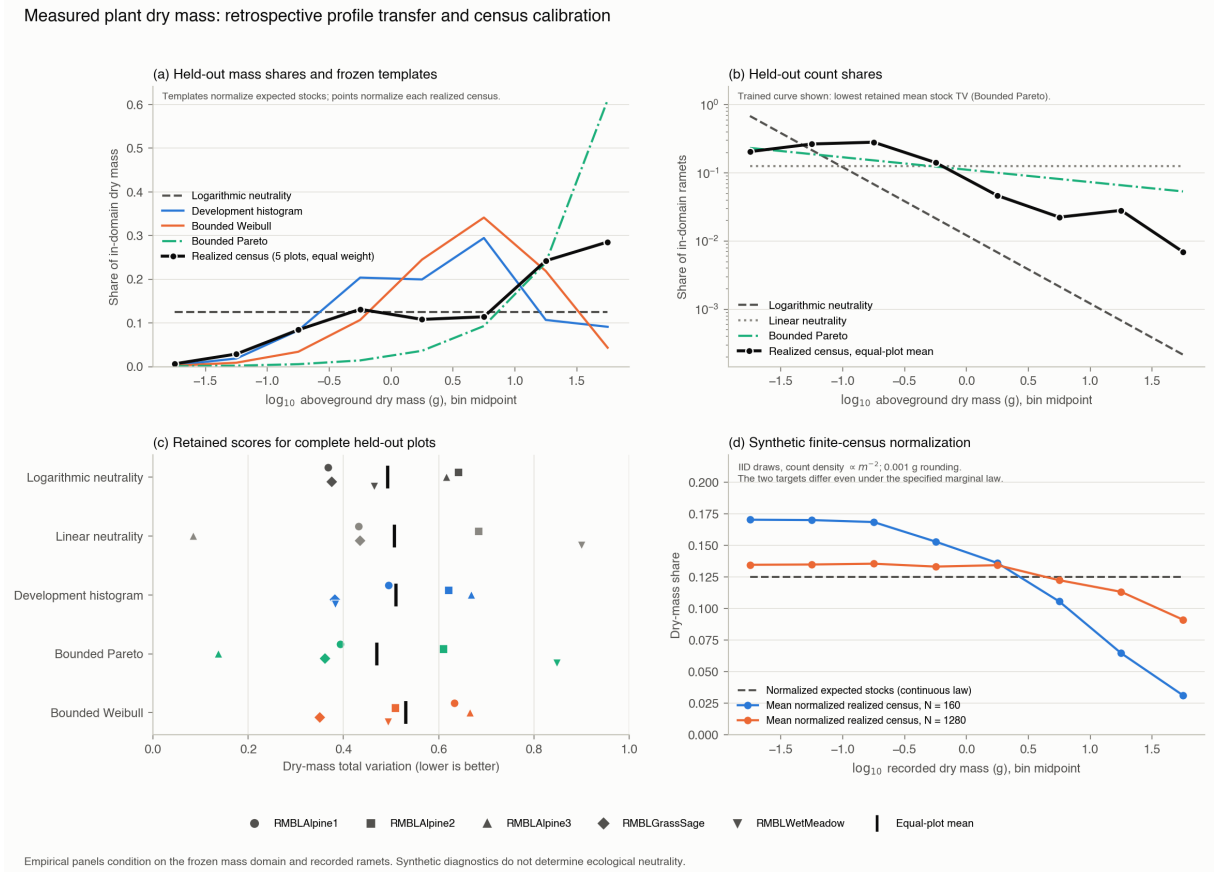


Figure 5. Directly weighed plant profiles and synthetic observation limits. (a) Equal-plot mean Colorado biomass shares and transferred stock templates. (b) Count shares and retained count forecasts. (c) Each plot’s biomass discrepancy and the five means. (d) Mean normalized census shares under iid logarithmic neutrality, compared with normalized expected stocks. All panels read retained outputs; `report_plant_biomass_profile.py` performs presentation only. Prior raw exposure and excluded biomass are retained.

5.5 What the tests establish

The positive results concern conditional budget closure: a separately measured or geometric cost, declared shared resource conditions, and abundance or resource shares predicted from them. The runtime experiment and the algal lineages are the clearest cases; in the lineages the closure holds with cost proportional to volume, so the abundance gradient carries no information beyond geometry. A rule that goes beyond closure, by predicting how a restriction redistributes resource among coexisting classes, failed in the grazed, dynamically changing food webs tested. The plant study now observes coexisting class stocks directly, but it does not establish equal expected allocation: the flat template is not best for biomass or counts, and the ecological observation law is uncalibrated. No natural size-class study in Section 5 has independently identified neutral eligibility and a design sufficient for the expected-allocation verdict. That remains the decisive missing test.

6. Discussion

Orthopolity is advanced here as a general natural-law proposal about resource allocation. Its neutral expression is equal effective resource per physically specified comparison measure; its observed expression includes the effects of the system and medium. Effective resources can combine several constituents nonlinearly. A distribution that is not flat in one separately measured constituent can therefore reflect a composite allocation, a constraint, or both. These possibilities become explanations when their combination rules and consequences are specified. The law does not identify every visibly unequal distribution with failure of the underlying tendency.

The class-exchange construction provides an explicit model in which an unequal stationary profile contains active opposing contributions. Removing the constraint predicts recovery without resetting the state. The electromagnetic examples establish related physical realizations with different observables: nested throughput, disjoint shell energy, and thermal energy per mode. Their measures and equality conditions follow from their respective equations. Quantization restricts available mode energies; it is a different kind of constraint from an additive transport bias. The framework can encompass these mechanisms without asserting that their dynamics are identical.

The inverse direction is central to the proposed law's scientific use. An abundance profile constrains effective resource divided by its allocation distortion, conditional on the comparison measure. A Pareto exponent can identify its scaling even when the constituent resource combination is unknown. It does not uniquely determine that combination, but shared constitutive families and independently characterized environments can make the inverse problem identifiable. The constructed example in Section 2.3 recovers a nonlinear resource from two environments and predicts a third. City-size scaling supplies an observational motivation for applying this strategy to coupled urban requirements. No city resource combination has yet been measured or fitted here.

The empirical studies contribute different kinds of evidence. Separately calibrated costs and shared resource conditions support budget-closure predictions in the engineered allocation system and the size-selected algal lineages. The chemostat cost-ratio rule fails its declared redistribution predictions, and plant stock/count comparisons remain mixed. The FIRAS transformation preserves the published quantum-spectrum result and its reconstruction assumptions. The natural size-class studies do not yet identify a neutral regime and its effective resource with sufficient observation information for a general expected-allocation verdict. A nonlinear composite hypothesis opens further research; it does not retrospectively change these studies' frozen predictions or outcomes.

The proposal builds on established work. Reversible transport and equipartition supply mechanisms and limiting cases; ecological size-spectrum models already connect resource use and abundance (Cuesta, Delius and Law, 2018; Arranz et al., 2022). Independent metabolic measurements already support cost-to-capacity predictions (Marshall et al., 2022). Proportional-growth models address city-size scaling (Gabaix, 1999). The contribution sought from orthopolity is a general allocation account that connects these results, organizes their constraints, and generates further forward and inverse predictions. These relationships require explicit derivations and comparisons, not an assumption that naming a common pattern establishes a common microscopic mechanism.

The next empirical advance should connect a physically interpretable effective-resource family to observations across conditions. Specify the object partition, comparison measure, constituent requirements and budgets, and observable target; use a calibration subset to infer the combination and its uncertainty; then predict another distribution or a response to a specified change. Variation in resource requirements must distinguish the candidate combinations, and the observation model must account for dependence and finite samples. A deeper microscopic mechanism can strengthen the explanation, but independent predictive success can also support an allocation law. This programme tests and extends the scope of the proposed law while using distributions to learn about the resources through which natural systems are organized.

7. Conclusion

We propose orthopolity as a general natural law: Nature tends to distribute resources equally among concentrations of those resources. The law’s expression depends on the system’s construction, medium, resource interactions and comparison measure. Its operative resource may be a nonlinear combination of several constituents. The framework therefore has both a forward use, predicting allocation from a resource model, and an inverse use, inferring effective resource structure from observed distributions.

The neutral allocation relation gives inverse-cost abundance under a declared measure. The class-exchange model supplies a mechanism, a constrained unequal equilibrium with opposing contributions, and quantitative recovery. Electromagnetic transport and thermal radiation supply worked physical realizations and explained departures. The retained observational studies contribute conditional successes, failed predictions and unresolved questions with their original scope preserved. A Pareto profile supplies a conditional constraint on effective-resource scaling; additional environments can identify and test candidate nonlinear combinations.

Together these results provide the proposed law’s mathematical and inferential foundation and several physical and empirical connections. They establish the stated conditional results rather than a universal identification of resources or mechanisms. The central research task is now to determine effective resources and constraints in further systems and test the new consequences that follow from the common allocation law.

Data and code availability

Code, frozen inputs, configurations, and generated results are in this repository. The make all target verifies input checksums, runs tests, and regenerates the empirical analyses; see the README for setup. Pilot estimates originate in results.json; distribution diagnostics in gof.json; aquatic median, compatibility fractions, and ocean intervals in independent.json. Remaining JSON files contain explicitly exploratory diagnostics. The studies of Section 5 are registered in an append-only run registry with their inputs, frozen protocols, archived algorithms and outputs. The class-exchange demonstration and retrospective FIRAS calculation are also registered; their

reports give offline reproduction commands and interpretation limits. Plant sources are pinned to author commit defccc3dcbbbf3ba57ff1572377de88fba83ff7f, with acquisition checksums and complete inclusion ledgers. The registry-verify target audits the registry offline, and each study report gives its replay command; source errata found during those audits are recorded there. Source-specific conditions are recorded in data/NOTICE.md; there is no blanket licence for all inputs. The supplied 2017 manuscript was inspected privately and is not redistributed.

References

- Arranz, I., Fournier, B., Lester, N. P., Shuter, B. J., and Peres-Neto, P. R. (2022). Species compositions mediate biomass conservation: The case of lake fish communities. *Ecology*, 103, e3608. <https://doi.org/10.1002/ecy.3608>.
- Boyd, S., and Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press. Sections 4.7 and 5.6. <https://web.stanford.edu/~boyd/cvxbook/>.
- Clauset, A., Shalizi, C. R., and Newman, M. E. J. (2009). Power-law distributions in empirical data. *SIAM Review*, 51, 661–703. <https://doi.org/10.1137/070710111>.
- Condon, J. J., and Ransom, S. M. (2016). *Essential Radio Astronomy*. Princeton University Press. Chapter 2, radiation and absorption. <https://www.cv.nrao.edu/~sransom/web/Ch2.html>.
- Cuesta, J. A., Delius, G. W., and Law, R. (2018). Sheldon spectrum and the plankton paradox: two sides of the same coin—a trait-based plankton size-spectrum model. *Journal of Mathematical Biology*, 76, 67–96. <https://arxiv.org/abs/1607.04158>.
- Damuth, J. (1981). Population density and body size in mammals. *Nature*, 290, 699–700. <https://doi.org/10.1038/290699a0>.
- Dillon, K. T., et al. (2019). On the relationships between size and abundance in plants: beyond forest communities. *Ecosphere*, 10, e02856. <https://doi.org/10.1002/ecs2.2856>.
- Edwards, A. M., Robinson, J. P. W., Plank, M. J., Baum, J. K., and Blanchard, J. L. (2017). Testing and recommending methods for fitting size spectra to data. *Methods in Ecology and Evolution*, 8, 57–67. <https://doi.org/10.1111/2041-210X.12641>.
- Edwards, A. M., Robinson, J. P. W., Blanchard, J. L., Baum, J. K., and Plank, M. J. (2020). Accounting for the bin structure of data removes bias when fitting size spectra. *Marine Ecology Progress Series*, 636, 19–33. <https://doi.org/10.3354/meps13230>.
- Eeckhout, J. (2004). Gibrat’s law for (all) cities. *American Economic Review*, 94(5), 1429–1451. <https://doi.org/10.1257/0002828043052303>.
- Ersoy, Z., et al. (2025). GLOSSAQUA: A global dataset of size spectra across aquatic ecosystems. *Ecology*, 106, e70050. <https://doi.org/10.1002/ecy.70050>.
- Fabbri, R., and Oliveira Jr., O. N. (2017). *A simple model that explains why inequality is ubiquitous*. Supplied manuscript dated 17 March 2017, 13 pp.; publication status unverified.
- Fabbri, R. (2024). The Orthopolity cosmological principle and the Natural distribution law. Author’s essay, 14 August. <https://ttm.github.io/2024/08/14/power.html>.
- Fixsen, D. J., Cheng, E. S., Gales, J. M., Mather, J. C., Shafer, R. A., and Wright, E. L. (1996). The cosmic microwave background spectrum from the full COBE FIRAS data set. *The Astrophysical Journal*, 473, 576. <https://doi.org/10.1086/178173>. Source article: <https://arxiv.org/abs/astro-ph/9605054>.
- Gabaix, X. (1999). Zipf’s law for cities: An explanation. *The Quarterly Journal of Economics*, 114(3), 739–767. <https://doi.org/10.1162/003355399556133>.
- Gaedke, U. (1993). Ecosystem analysis based on biomass size distributions: A case study of a plankton community in a large lake. *Limnology and Oceanography*, 38, 112–127. <https://doi.org/10.4319/lo.1993.38.1.0112>.

- Harcourt, R., Garcia, N. S., and Martiny, A. C. (2024). *Synechococcus* batch culture data (cell quotas and ratios (C, N, P), size, and diameter) from laboratory experiments in 2021 to 2022 with related isolates cultured across a range of temperatures. BCO-DMO dataset 926311, version 1. <https://doi.org/10.26008/1912/bco-dmo.926311.1>.
- Hatton, I. A., Heneghan, R. F., Bar-On, Y. M., and Galbraith, E. D. (2021). The global ocean size spectrum from bacteria to whales. *Science Advances*, 7, eabh3732. <https://doi.org/10.1126/sciadv.abh3732>.
- Haus, H. A., and Melcher, J. R. (1989). *Electromagnetic Fields and Energy*. Prentice-Hall. Section 11.2, Poynting's theorem. https://web.mit.edu/6.013_book/www/chapter11/11.2.html.
- Ioannides, Y. M., and Skouras, S. (2013). US city size distribution: Robustly Pareto, but only in the tail. *Journal of Urban Economics*, 73(1), 18–29. <https://doi.org/10.1016/j.jue.2012.06.005>.
- Jaynes, E. T. (1957). Information theory and statistical mechanics. *Physical Review*, 106, 620–630. <https://doi.org/10.1103/PhysRev.106.620>.
- Jaynes, E. T. (1968). Prior probabilities. *IEEE Transactions on Systems Science and Cybernetics*, 4(3), 227–241. <https://bayes.wustl.edu/etj/articles/prior.pdf>.
- Levin, D. A., and Peres, Y. (2017). *Markov Chains and Mixing Times*, second edition, with contributions by E. L. Wilmer. American Mathematical Society. <https://pages.uoregon.edu/dlevin/MARKOV/mcmt2e.pdf>.
- Malerba, M. E., Palacios, M. M., and Marshall, D. J. (2018). Do larger individuals cope with resource fluctuations better? An artificial selection approach. *Proceedings of the Royal Society B*, 285, 20181347. <https://doi.org/10.1098/rspb.2018.1347>. Data: <https://doi.org/10.5061/dryad.4mh47r7>.
- Marshall, D. J., et al. (2022). Long-term experimental evolution decouples size and production costs in *Escherichia coli*. *Proceedings of the National Academy of Sciences*, 119, e2200713119. <https://doi.org/10.1073/pnas.2200713119>.
- Newman, M. E. J. (2005). Power laws, Pareto distributions and Zipf's law. *Contemporary Physics*, 46, 323–351. <https://arxiv.org/abs/cond-mat/0412004>.
- NASA LAMBDA (accessed 6 October 2026). COBE FIRAS CMB monopole spectrum, version 1. Product description and reconstruction provenance. https://lambda.gsfc.nasa.gov/product/cobe/firas_monopole_spect.html.
- Planck, M. (1901). Ueber das Gesetz der Energieverteilung im Normalspectrum. *Annalen der Physik*, 309, 553–563. <https://doi.org/10.1002/andp.19013090310>.
- Perkins, D. M., et al. (2018). Bending the rules: exploitation of allochthonous resources by a top-predator modifies size-abundance scaling in stream food webs. *Ecology Letters*, 21, 1771–1780. <https://doi.org/10.1111/ele.13147>.
- Poynting, J. H. (1884). On the transfer of energy in the electromagnetic field. *Philosophical Transactions of the Royal Society of London*, 175, 343–361. <https://doi.org/10.1098/rstl.1884.0016>.
- Sheldon, R. W., Prakash, A., and Sutcliffe, W. H., Jr. (1972). The size distribution of particles in the ocean. *Limnology and Oceanography*, 17, 327–340. <https://doi.org/10.4319/lo.1972.17.3.0327>.
- Tong, D. (n.d.). *Statistical Physics*. University of Cambridge Part II Mathematical Tripos lecture notes, Sections 1.3, 2.2.1 and 3.2. Accessed 6 October 2026. <https://davidtong.org/pdfs/teaching/statistical-physics/statphys.pdf>.
- Visser, M. (2013). Zipf's law, power laws and maximum entropy. *New Journal of Physics*, 15, 043021. <https://arxiv.org/abs/1212.5567>.
- Wojcik, L. A. M., Pfennig, A., Flamm, S., Klauschie, T., Rosenbaum, B., Weithoff, G., and Gaedke, U. (2025). Top-down control and species composition influence nonlinearly the short-term response of experimental food webs to a nutrient pulse perturbation. *Proceedings of the Royal Society B*. <https://doi.org/10.1098/rspb.2025.1969>. Data: <https://doi.org/10.5061/dryad.51c59zwj5>.